

WASP-10b

R-1653

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_200728 · created 2026-08-02T08:28:21+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459058.8026 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.211 +/- 0.061
Transit depth (Rp/Rs)^2	4.45 +/- 2.58 [%]
Orbital Inclination (inc)	89.35 +/- 1.85
Airmass coefficient 1 (a1)	1.05 +/- 0.43
Airmass coefficient 2 (a2)	0.03 +/- 0.19
Scatter in the residuals of the lightcurve fit is	42.45 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.75
Transit Duration (day)	0.094 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.211 ± 0.061 vs 0.1592 (deviation 0.85σ)
Duration fit vs published	0.094 d vs 0.0946 d (deviation 0.05σ)
Mid-time O–C	-2.8 min
Depth SNR	3.5
Residual scatter	42.45 %

Light curve

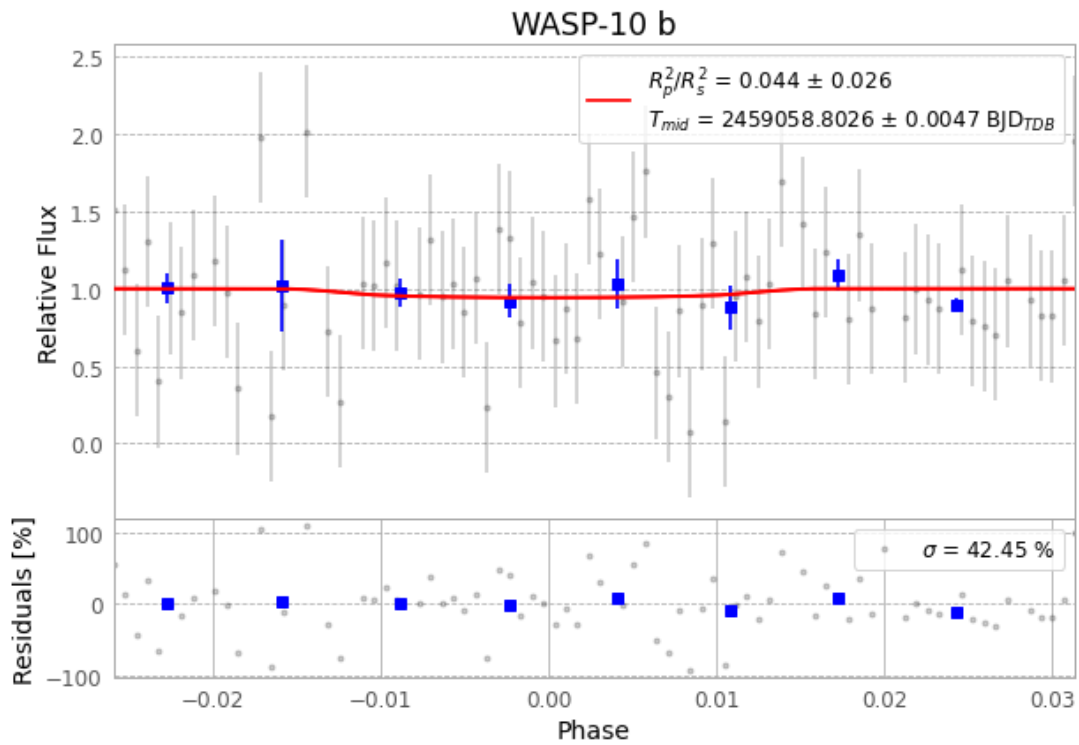


Figure 1 — FinalLightCurve_WASP-10 b_2020-07-27.png (SHA-256 cafb5f05c35e25f2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [608, 336], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d17c4c23769361b8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), WASP-10200728051512.FITS → WASP-10200728093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-200728.log (d1c36a023a65...), FinalLightCurve_WASP-10 b_2020-07-27.png (cafb5f05c35e...), FinalLightCurve_WASP-10 b_2020-07-27.csv (b397c41e7fd1...)

Compute resources

Wall time 295.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 08:28Z.